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Cognitive intelligence-enabled requirements elicitation for design optimisation of smart product-service system

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ABSTRACT

The rapid development of advanced technologies provides futuristic option to enhance product value with digital services, which the concept of Smart Product-Service System (SPSS) was proposed. SPSS focuses on usage data, the whole lifecycle, and the iterative optimisation. A large amount of data and context information can be collected during the usage stage, which can be used to elicit requirements to guide further design. How to realise the requirement elicitation with cognitive intelligence based on usage data is the key step for design optimisation. Therefore, this research proposes a cognitive intelligence-enabled requirements elicitation framework to help designers promote the design optimisation process. First, the digital twin-driven data collection architecture of SPSS is proposed by considering multi-sources and multi-stages. Second, the behaviour of SPSS is modelled based on Auto-Encoder which is constructed using deep learning to cognise SPSS performance. Third, the corresponding behaviour model is interpreted based on Shapley Addictive exPlanations (SHAP) to discover key factors with cognitive intelligence. Finally, the domain knowledge graph is used to generate the optimisation requirements for further design. To demonstrate the feasibility and advantages of the proposed framework, this study adopts the case study of wind turbines for illustration and verification.

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Smart product-service system (SPSS); cognitive intelligence; requirement elicitation; digital twin; model interpretation

1. Introduction

In recent years, the rapid advancement of Information and Communication Technologies (ICT) has facilitated widespread digital transformation, bridging the gap between the physical and virtual realms (Mishra and Deichmann 2016; Song and Cao 2017; Zheng, Wang, et al. 2019). Simultaneously, the growing demand for highly personalised products necessitates enterprises to diversify their business models beyond traditional product offerings (Qu et al.

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2016). To effectively adapt to this shift, enterprises are expanding their business models to provide not only physical products, but also personalised services bundled together as a holistic solution tailored to individual user requirements (Xiang et al. 2024). This transformative shift has given rise to a new business paradigm, Smart Product-Service Systems (SPSS) (Goedkoop 1999; Valencia et al. 2015). In the realm of SPSS, digital services are built upon a foundation consisting of Smart, Connected Products (SCPs) (Zheng, Chen, and Wang 2021). Capitalising on the connectivity of products and the real-time intelligence of services, SPSS represents a pivotal approach for enterprises to overcome the challenges of product homogeneity and better cater to the personalised requirements of users (Zheng et al. 2018; Zheng, Chen, and Shang 2019). Due to the increased complexity and functionality inherent to SPSS compared to single products, the design methods and associated mechanisms of SPSS have become a prominent focal point of research.

SPSS advocates the responsibility of enterprises to integrate the services and products in the whole lifecycle, bringing users a better experience and improving their economic benefits simultaneously (Fagnoli et al. 2018). Traditional design methods often require designers to collect many requirements and design a new generation of products to meet market needs (Freddi and Salmon 2019). This leads to a lag in responding to user needs, which has a negative impact on both users and enterprises. However, it is important to realise fast correspondence to improve the total performance of SPSS. From the perspective of the SPSS lifecycle, the value of SPSS could be increased with design optimisation in the usage stage. The smartness of SPSS reflects that a significant quantity of data generated during the usage stage can be collected and analysed in real-time to optimise the system performance and improve user satisfaction (Chen et al. 2020; Kühlenkötter et al. 2017).

The process of requirements elicitation plays a crucial role in improving the support provided by the data generated during the usage stage in optimising the design of SPSS. The large volume of data generated during the usage stage is vital in driving the innovation of SPSS's design and plays a role in determining its ultimate performance (Zhang et al. 2020). Considering requirements are dynamic, the critical element to requirement identification lies in finding users' pain points in the usage stage to help enterprises offer better experiences. Requirements elicitation provides significant guidance for the following-up optimal design of function, behaviour, structure, control, intelligence, and performance of SPSS, as well as the efficient commissioning of the corresponding manufacturing system (Leng et al. 2021a; Leng et al. 2021b; Leng et al. 2023). At the same time, service requirements may emerge during the usage stage. It is crucial to utilise the features of SPSS and leverage the data gathered through it to identify individualised service requirements of users and enhance the product's overall value (Chang et al. 2019; Masood, Cherifi, and Moalla 2020). Considering the features of complex structure and robust interactivity of SPSS, the existing representation of service requirements should be adapted to these features. Convenient approaches often confuse designers in the discovery and management of requirements in a rapidly changing environment (Song 2017). Moreover, the inefficiency in requirement elicitation and response sometimes leads to barriers between usage and design stages. And the requirement elicitation process needs the support of knowledge to enhance the rationality of design optimisation decisions.

For these issues, considering cognitive intelligence provides a practical pathway for machines to gain intelligence within the integration of data processing and knowledge management (Chen, Herrera, and Hwang 2018). Cognitive intelligence is triggered by the

Table 1. Three main stages in the development of PSS concept.

Stage	Definition	References
Internet-based PSS	Internet, web enabled and wireless technologies help customers to have more and more options in front of them at an instant of a click. Their access to more suppliers and products through Internet unlimits their power of purchase, and empower them to ask better products at a minimum cost with extended service additions.	Lee (2003)
IoT-enabled PSS	The concept of the product-service system (PSS) has been increasingly used as a new business model in implementing advanced technologies including RFID	Baines et al. (2007), Michael et al. (2010)
Smart PSS (SPSS)	An IT-driven value co-creation business strategy consisting of various stakeholders as the players, intelligent systems as the infrastructure, smart, connected products as the media and tools, and their generated e-services as the key values delivered that continuously strives to meet individual customer needs in a sustainable manner	Wiesner, Marilungo, and Thoben (2017), Zheng et al. (2018)

data collection, and the result is reflected in the design decisions. It can be seen that the advantages of cognitive intelligence fit the dilemmas of SPSS design optimisation. Cognitive intelligence-enabled SPSS design optimisation could improve the system performance from the following three dimensions: (1) Handle massive real-time data; (2) Application domain knowledge; (3) Reasoning process based on data-knowledge fusion (Chen, Herrera, and Hwang 2018; Wang, Liu, and Zheng 2022).

To deal with the complexity of coupling products and services in SPSS, strengthen the combination of the usage and design stage, and improve the efficiency of design optimisation, this paper proposes a cognitive intelligence-enabled requirement elicitation framework to help designers in the process of SPSS design optimisation. This paper is organised as follows: Section 2 reviews the related literature on SPSS and requirements elicitation within the SPSS field. Section 3 illustrates the proposed cognitive intelligence-enabled SPSS optimisation design approach, concluding: digital twin-based data collection, behaviour modelling-based SPSS cognition, SHAP-based SPSS cognition intelligence, and knowledge-aided requirement generation. Section 4 presents a case study to verify the proposed framework for design optimisation requirement elicitation of SPSS. Section 5 concludes this paper and discusses future work.

2. Literature review

2.1. SPSS definitions and design methods

After being first proposed by Goedkoop in 1999, Product-Service System (PSS), has mainly gone through three main stages with the development of information technologies, which is shown in Table 1. According to the definition of SPSS, the significant research object of SPSS is smart, connected products (SCP) (Zheng, Wang, et al. 2019), which is composed of physical entities, intelligent components, and interconnected components, with functions of monitoring, optimisation, decision-making, and controlling (Cong et al. 2020; Wang and Chin 2017). In conclusion, SPSS is a way to enhance the personalised solution development in an intelligent way and optimise the overall performance with product-service solutions.

Based on the significant differences between SPSS and traditional product, the design methods of SPSS vary from classical product design methods. According to relevant research, SPSS design process could be divided into two aspects: data-driven platform

design process and value-driven co-creation process. Data-driven platform design process relies on the valuable data collected from SCP (Li et al. 2021; Machchhar et al. 2022; Wang et al. 2019) and using the platform to realise data analysis and service recommendation. Valencia et al. proposed the early concept to connect intelligent products and services using smart connected technologies (Valencia et al. 2015). Liu et al. regard SPSS as a platform composed of intelligent products and intelligent services, and the brilliant service ecosystem is composed of competent services, further strengthening the performance of SPSS (Liu et al. 2018). Zheng et al. built a platform-based, data-driven framework for SPSS service innovation and validated it on wearable devices (Zheng et al. 2018). Li et al. proposed the combination of knowledge graphs and conceptual knowledge models to assist the SPSS design process and automatically generate SPSS innovative design schemes under the trigger of individual requirements (Li et al. 2020). To improve the overall performance of data-driven process, technologies such as virtual reality (Liu and Yang 2022), digital twin (Tao et al. 2018), etc., have also shown in some related research. Value-driven co-creation design methods illustrate the balance of different stakeholders' benefits in SPSS development process (Cui et al. 2023). Kuhlentötter et al. viewed SPSS as a highly digital value co-creation system characterised by high complexity, dynamics, and interconnectivity (Kuhlentötter et al. 2017). Lee et al. proposed a service design approach driven by overall customer requirements with traceability logic, using ladder theory, ontology design theory, TRIZ theory, and QFD technology to effectively process design knowledge and generate design solutions in innovative conceptualisation (Lee, Chen, and Lee 2020; Lee, Chen, and Trappey 2019). Wang et al. proposed combining TRIZ theory and correlation analysis to discover product deficiencies and generate product concept design (Wang 2015; Wang and Chin 2017).

2.2. Requirement elicitation method

Requirement elicitation is always the first step in the design process of any entity or software. High-quality requirement engineering can help project stakeholders increase their understanding of the details of complex product features and enhance the relationship between the design objects and their market. Over the past few decades, numerous methods have been suggested for achieving requirements elicitation for physical entities. From the source of the requirement elicitation, it could be divided into twin categories: user-generated and design object-generated. According to the data type perspective, it could be divided into text and numeric type. According to the above classification rules, requirement elicitation methods could be classified into four categories, which is shown in Figure 1.

The requirement elicitation methods based on user-related sources mainly relies on user comments, interviews, personas, etc. Wang et al. proposed a method for online reviews-based requirements extraction and design proposals generation (Wang et al. 2024). Zhang et al. proposed a Grey-Markov model of user requirements prediction based on online reviews for enterprises (Zhang et al. 2023). Considering that multiple user requirements are essential for enterprises to make decisions, Lai et al. elicited users' implicit requirements regarding product function and performance through text mining of product's online reviews and used graph theory to help market analysis and consumer behaviour modelling (Lai et al. 2019). Requirement elicitation methods based on user-generated numeric data

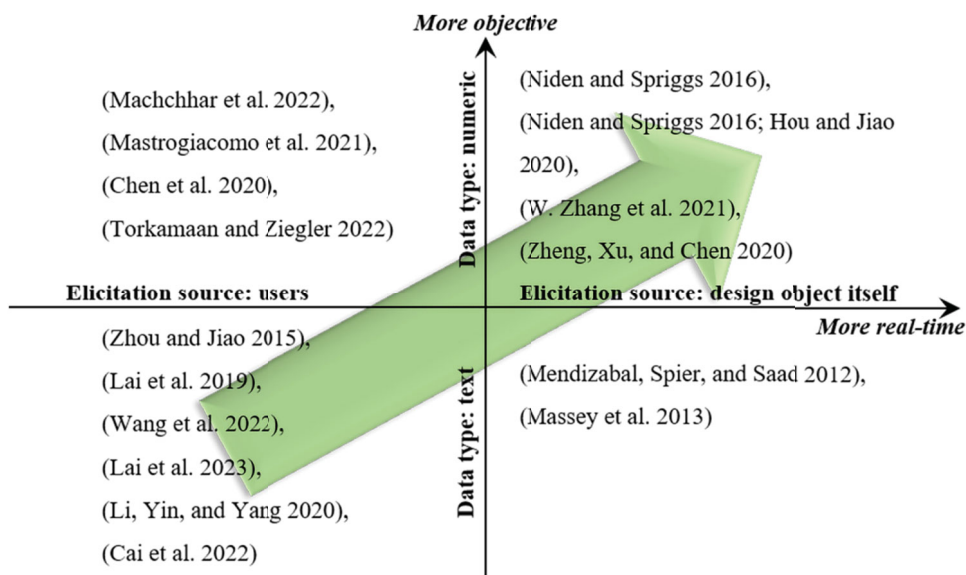


Figure 1. Requirement elicitation methods in four categories.

mainly through analysing users' behaviour data. Wang et al. proposed a data-driven graph-based SPSS requirement elicitation framework based on user-generated data collected from sensors, and a requirement graph is constructed to find the high-reliability requirements with the help of the random walk algorithm (Machchhar et al. 2022). Mastrogiacomio et al. suggested a way to identify the quality determinants of PSS from unstructured User-Generated Content (UGC) through text mining in response to the ever-changing demands of users (Mastrogiacomio et al. 2021). This framework aims to match key activity elements with smart capabilities to elicit service requirements throughout the user activity cycle. Chen et al. suggested a hybrid framework for identifying and prioritising service requirements based on user activities (Chen et al. 2020). Barré et al. proposed a persona-based requirement elicitation method for multidisciplinary teams (Barré, Buisine, and Aoussat 2017). Considering that multiple stakeholders participate in the use process, conflicts may exist between massive requirements (Song 2017). To make requirement conflicts resolution, Marina et al. proposed a requirements data model for PSS requirements, which could realise conflict detection and resolution (Berkovich et al. 2014).

Considering user-generated information may contain uncertainty due to different users' preference, more and more research has paid attention on requirement elicitation based on data generated from the design object itself. With the rapid development of IoT and CPS, data are reshaping most functions from conventional products and are full of every stage of the product lifecycle (Hou and Jiao 2020; Niden and Spriggs 2016). Zhang et al. suggested an innovative method for optimising personalised product design through data-driven inverse design optimisation (Zhang et al. 2021). This involves incorporating forward design and inverse problem-solving into a cohesive framework for data analysis, resulting in a closed-loop decision-making process. Zheng et al. introduced a data-driven cyber-physical approach for developing personalised smart connected products (SCP) in a cloud-based setting. This method involves utilising co-development toolkits that possess

both smartness and connectedness to enable interaction processes (Zheng, Xu, and Chen 2020). Requirement elicitation methods based on text from the design object itself mainly on the basis of running log, policy documents, etc. Mendizabal et al. presented a practical approach for requirements elicitation and prioritisation based on realistic user behaviours log files (Mendizabal, Spier, and Saad 2012).

2.3. Cognitive intelligence

With the application of information technologies such as SCPs, IoTs, artificial intelligence, etc., the efficiency of lifecycle design has been developed (Niu, Wang, and Qin 2021). Through the enhancement in smart abilities containing context awareness, status monitoring, real-time analysis, cognition intelligence could be implemented to support the design process of SPSS. SPSS handles physical products and digital services as a bundle that lays the foundation for cognitive intelligence. Moreover, with the infusion of knowledge, the interconnection between data and knowledge could enhance the efficiency and accuracy of the cognition process towards the working stage in SPSS lifecycle (Mourtzis 2021).

Cognition is a concept derived from the human's physiological process. Human's cognition is triggered from the sensation of the environment and the information collection, and the end result is reflected in the final decisions and physical actions (Stillings 1995). Strube defined cognition as *a class of advanced control mechanisms that allow for sophisticated adaption to changing needs (e.g. learning and planning) through computations operating on mental representations* (Strube 2001). According to related theories, the cognition science is always based on the collection of knowledge, the ability of learning, and the nature of motivation and engagement (Agbozo et al. 2022). Reference to human's cognitive processes, SPSS could realise the cognition process with domain knowledge acquisition through NLP toolkits and knowledge management through the knowledge graph. For SPSS, the ability of learning is reflected in learning important information such as patterns and features from massive data through deep learning methods. And the motivation for SPSS is to make better performance of the system.

Nowadays, more and more researchers have paid attention to the combination of cognitive intelligence and industrial scenes. Agbozo et al. proposed the concept of cognitive intelligence-enabled manufacturing (CoIMS) and analysed the fundamentals and enabling technologies of CoIMS (Agbozo et al. 2022). Zheng et al. proposed a self-X cognitive manufacturing network with the combination of knowledge management and multi-agent reinforcement learning method (Zheng et al. 2021). Faruque et al. studied the cognitive digital twin which could allow enterprises to exploit implicit knowledge drawn from manufacturing systems and enhance the intelligent ability of manufacturing enterprises (Faruque et al. 2021).

2.4. Motivation

Although existing requirement elicitation methods can realise the process with technologies such as text mining, data analysis, etc. However, there still exists some weakness in the design optimisation requirements elicitation process of SPSS. The complexity of the system structure brings a lot of uncertainty to the process of design optimisation. Traditional requirements elicitation methods have a lag between the usage stage and the design

stage. The mechanism for the SPSS design optimisation to respond quickly and accurately to surrounding information needs to be constructed, which could enhance the connection across stages. Moreover, facing the optimisation requirements generated from the usage stage, multiple paths should be considered for SPSS to fulfil them, mainly containing service recommendation and product upgrade. This will improve the low efficiency of SPSS design optimisation. To enhance SPSS optimisation, there exists weakness in linking system information with human understanding, which results in poor interpretability of the optimisation solution. Introducing cognitive intelligence serves as the crucial facilitator to bridge the semantic gap between humans and SPSS.

Therefore, how to realise the cognitive process of SPSS requirement elicitation with efficient data analysis has great significance for the design process. Moreover, considering the real-time property of SPSS, it is valuable to study the requirement elicitation methods upon usage data to support the design optimisation process. Motivated by the above summary, a cognitive intelligence-enabled design optimisation requirement elicitation framework of SPSS is introduced in the following.

3. Cognitive intelligence-enable requirement elicitation framework for SPSS

A cognitive intelligence-enabled SPSS requirement elicitation framework is constructed to support the design optimisation process, which is shown in Figure 2. The Framework is composed of four parts, which will be introduced in sections 3.1–3.4. According to Section 3.1, data are collected from SPSS. Then, the behaviour model is trained with deep learning, composed of the cognition process in Section 3.2. Section 3.3 introduces the model interpretation-based SPSS cognition intelligence, and model interpretable methods are adopted to reveal the key factors reflected behind usage data. Finally, optimisation requirements are generated using the domain knowledge graph in Section 3.4, which could support designers in the design optimisation process.

3.1. Digital twin-based data collection

Out of concern for the enhancement of advanced intelligent technologies, it is necessary and capable to focus on the whole lifecycle of SPSS. SPSS lifecycle includes design, manufacturing, distribution/logistics, usage, and end-of-life stages (Stark 2022; Zheng, Lin, et al. 2019). Different stages correspond to various stakeholders, including executors, beneficiaries, etc., which could vary widely from individuals to enterprises. In the usage stage of SPSS lifecycle, a large number of valuable data could be generated, which is related to multi-stakeholders containing users, after-sale service providers, etc.

Moreover, the performance of SPSS in the usage stage always corresponds with the benefits of enterprises and the development of specific industries. However, the performance of SPSS in this stage could be easily influenced by different contexts and operation methods. Therefore, it is especially significant to dig out and convert much more valuable information from the usage stage of SPSS, which could play a valuable role in the design, manufacturing, and other stages of SPSS's lifecycle.

Data generated from SPSS usage stage work has features like multi-source, heterogeneous, massive, etc., which needs an efficient and comprehensive way to deal with. As an advanced digitisation technology, digital twin (DT) has integrated the Internet of Things,

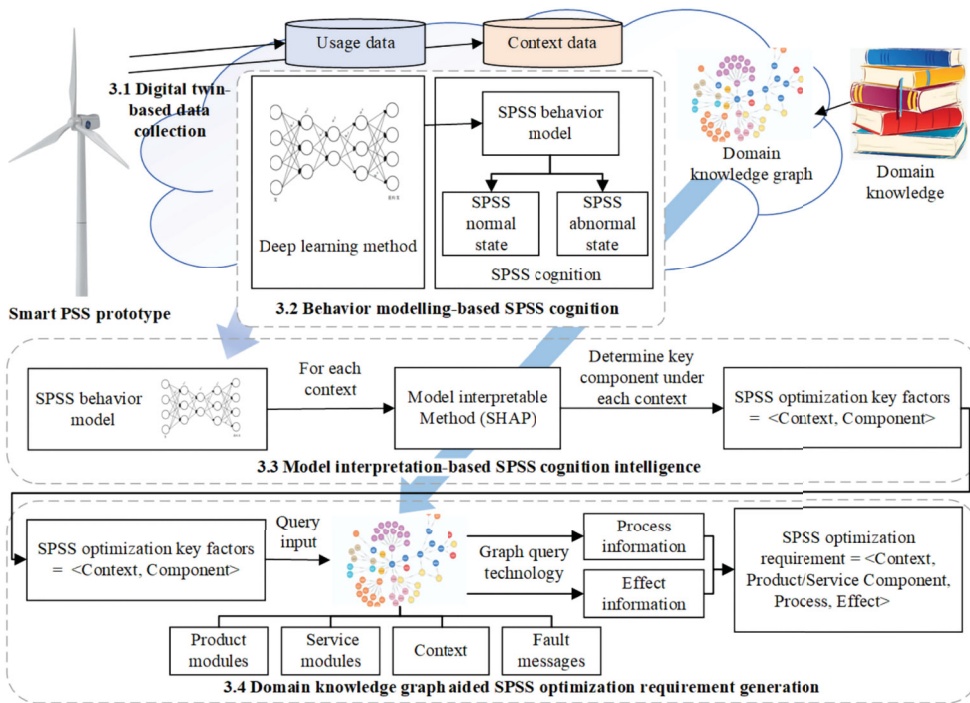


Figure 2. Proposed research framework.

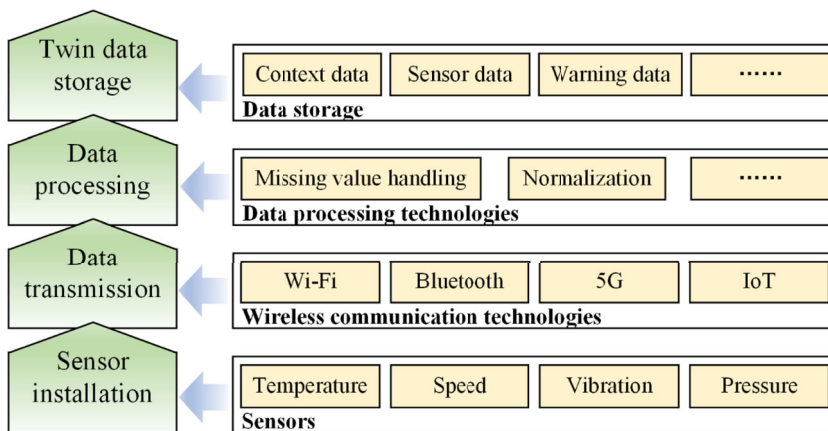


Figure 3. Digital twin-based data collection process.

big data, artificial intelligence, etc. The basic principle of DT is to realise monitoring, controlling, predicting, and optimisation by bridging the physical entities and the digital model. A DT-based data collection process is constructed to realise the data collection in the SPSS usage stage, as shown in Figure 3. The DT-based data collection is composed of sensor installation, data transmission, data processing, and data storage.

The digital twin-based data collection process is on the basis of different kinds of sensors which are also corresponding to SCPs in SPSS. With sensors in SCPs, different types

of data could be collected. Within wireless communication technologies, collected data could be transported into the twin data platform. Upon the twin data platform, many operations could be added to improve the data qualities such as data processing and storage as context data, sensor data, and warning data.

3.2. Behaviour modelling-based SPSS cognition

The target of SPSS cognition is to depict the laws and patterns of SPSS during the usage stage, which could be realised through behaviour modelling. Behaviour modelling refers to determining a monitoring indicator based on several characteristic variables of SPSS. With the well-learned behaviour model, the current status of SPSS could be decided normal or abnormal with comparisons. To realise behaviour modelling, there exist some methods based on traditional machine learning, which rely on feature engineering for a considerable proportion. In feature engineering, various feature processing methods and sufficient professional background knowledge are required to do the feature selection process. Therefore, for SPSS, feature selection will take much time cause the bigger data volume than traditional product and will take poor generalisation ability to the cognitive process. In recent years, deep learning has attracted great attention due to its powerful ability in adaptive feature extraction. In the field of behaviour modelling, it has also been widely used. Compared with machine learning, the biggest advantage of deep learning is that it can fully learn the intrinsic information of SPSS through multi-layer neural networks and does not require a lot of manpower or knowledge for manual feature extraction and screening.

Auto-Encoder (AE) is a typical deep learning model for SPSS behaviour modelling. It has the advantages of simple construction, stacking multiple layers, and is supported by neuroscience. It can extract low-dimensional nonlinear features that represent high-dimensional data. AE is a neural network model which is composed with encoder and decoder. The structural characteristics of AE are suitable for extracting latent features of massive data. With the encoding network, the original input data could be mapped to a lower-dimensional feature space to extract features automatically. Subsequently, it proceeds to the decoding process, where the low-dimensional data is reconstructed through the decoding network, resulting in the output of the reconstructed data. The reconstructed error is used to describe the similarity between the original and reconstructed data. The weight parameters at all AE networks are initialised in the unsupervised pre-training phase. AE uses back-propagation algorithms to update model parameters, which belongs to one kind of unsupervised learning.

AE could learn the SPSS working pattern after pre-training the AE networks with SPSS normal behaviour data. Under this case, the reconstruction error could be found in a normal range. When behaviour data collected from abnormal contexts, the reconstruction error would exceed the normal range. That's the mechanism using AE as the behaviour modelling method and using reconstruction error as the index for cognising whether the working status is abnormal or not. The cognition process with AE is shown in Figure 4.

The function of the encoder part is to reduce the dimensionality of the input data and compress the massive information. The number of neuron nodes in the hidden layer must be less than the dimensions of the input data to extract its potential features. In the

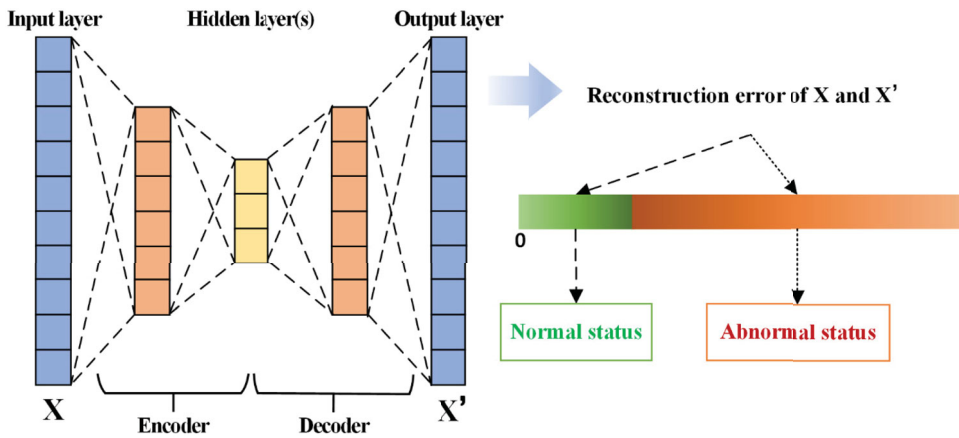


Figure 4. Behaviour modelling-based SPSS cognition.

encoding process, the data processing process is described by Equation (1).

$$h(x) = \sigma(W_1X + b_1) \quad (1)$$

In Equation (1), $X = \langle x_1, x_2, \dots, x_n \rangle^T$ represents the vector from the input layer. $W_1 \in m \times n$ is the weight matrix between the input layer and the hidden layer. $b_1 \in R^m$ represents the bias vector in the transfer process. σ is the activation function of AE, and $h(x) \in R^m$ means the value in the hidden layer.

As for the decoding stage, the output x is calculated through Equation (2).

$$X' = \sigma(W_2h + b_2) \quad (2)$$

In Equation (1), $X' = \langle x'_1, x'_2, \dots, x'_n \rangle^T$ represents the output vector in the output layer. $W_2 \in m \times n$ is the weight matrix between the hidden layer and the output layer. $b_2 \in R^n$ represents the bias vector in the transfer process.

AE can be viewed as a mapping process. During the forward propagation stage of AE, if the weights of the connections between elements yield definite results, the model should produce specific outputs. To ensure that the AE model provides accurate results, it must undergo training iterations during which its parameters are updated to achieve an excellent fit. These processes constitute the back-propagation of AE. Back-propagation aims to minimise the reconstructed error between the model's output values and the actual values, typically quantified by loss functions such as mean absolute error (MAE), mean squared error (MSE), cross-entropy, and others.

When the loss function is selected, the back-propagation process of AE can be regarded as a process that takes the weights between layers as variables and minimises the loss function by gradient descent. The formula for gradient descent is shown as follows, which is represented as an update of the weights in the AE model.

$$\theta_i = \theta_i - \alpha \frac{\partial}{\partial \theta_i} J(\theta_0, \theta_1, \dots, \theta_n) \quad (3)$$

α is the learning rate, which represents the learning speed. If it is too low or too high, time and precision will be influenced while training. $\frac{\partial}{\partial \theta_i} J(\theta_0, \theta_1, \dots, \theta_n)$ is the partial derivative of the loss function concerning θ_i .

With the well-trained AE model, SPSS behaviour modelling could be realised and the reconstruction error is set to make the cognition function. Regarding the SPSS abnormal operation situation, there will be changes in the relationships between the corresponding parameters due to the system dynamics theory, and the abnormal state at that moment could be judged accordingly.

3.3. Model interpretation-based SPSS cognition intelligence

When the key structural parameters of the SPSS abnormal behaviour are input into the well-trained SPSS normal behaviour model, the reconstruction error that deviates from the normal range will be calculated out. In this way, the abnormal state could be cognised. Moreover, to improve the accuracy of cognition, it is worth considering that learn SPSS behaviour modelling based on different contexts. To support the design process, it is vital to understand the mechanism of SPSS behaviour and extracted valuable information for designers. Therefore, exploring how key components cause a significant impact on the performance of SPSS in different critical contexts through data science methods is worth paying attention to, which could be further transformed into related optimisation key factors for requirements elicitation.

To elucidate the black-box nature of the deep learning model, shedding light on the individual feature impacts and cognising the behaviour of SPSS, model interpretability methods are recommended in this process. In the realm of model interpretability for deep learning applications, the selection of an appropriate method is of paramount importance to ensure robust and insightful insights into model predictions. The Shapley Additive exPlanations (SHAP) framework stands out as a compelling choice due to its ability to provide both global and local interpretability. The versatility and comprehensiveness of SHAP make it a well-suited and indispensable tool for unravelling the complexities inherent in deep learning models, thereby facilitating a more informed and interpretable analytical workflow. SHAP values offer a rigorous mathematical foundation, grounded in cooperative game theory, enabling a nuanced understanding of the contribution of each feature to a given result. Moreover, SHAP's capacity to handle complex model architectures, including deep learning networks, aligns seamlessly with the intricacies of the AE model.

With cognitive intelligence based on the SHAP method, SPSS behaviour model can be interpreted, and the key component with the strongest correlation under each context can be excavated. The corresponding output of this process is structured to a specific form named SPSS optimisation key factors, which is constructed into an associated vector of context and component in the following format:

$$\text{SPSS optimization key factors} = \langle \text{Context, Component} \rangle \quad (4)$$

Considering SPSS key factors are not detailed enough to guide SPSS design optimisation process, the SPSS key factors need to be transformed and generated into SPSS design requirements with reference significance in using knowledge graphs of the relevant domain fields.

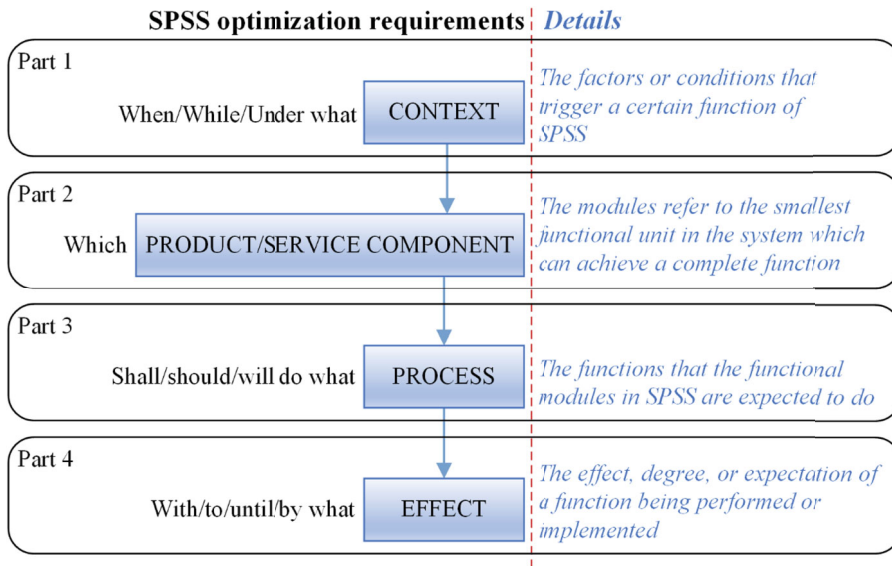


Figure 5. SPSS requirements characterisation template based on improved RUPP's template.

3.4. Domain knowledge graph aided SPSS optimisation requirement generation

RUPP's requirements characterisation template is one of the mainstream requirements characterisation templates, in which the requirements are characterised as *When / While / Under what CONTEXT, system COMPONENTS shall / should / will do what PROCESS*. Considering the smart ability of SPSS and better support designer's decision process, the above template is further improved as follows: *When/While/Under what CONTEXT, SYSTEM COMPONENTS shall/should/will do what PROCESS with/to/until/by what EFFECT*, which is shown in Figure 5.

Among the templates, context refers to the factors or conditions that trigger a specific function of SPSS, such as the indication of the sensor or a particular physiological parameter of the human body. System components are the functional units that can achieve a complete function. Components in SPSS include both product-related modules and service-related modules. Process refers to the tasks that the available modules in SPSS do. The effect refers to the mode, degree, or expectation of a function being performed or implemented. For example, following the mentioned template, one of the SPSS requirements could be described as *When the user's body temperature is higher than 37.3 degrees Celsius, the alarm device emits an alarm signal of fever*. The context is that *user's body temperature is higher than 37.3 degrees Celsius*, and the system component is *the alarm device* (product module) mentioned in the text. The process corresponds to the *emits an alarm signal* in this sentence, which is the expected function provided by the module. *As soon as possible* is the performance requirement of the function execution, which is the content of the effect. Through the above settings, an SPSS requirement description vector is proposed, namely:

$$\text{SPSS optimization requirement} = \langle \text{Context}, \text{Product/Service Component}, \text{Process}, \text{Effect} \rangle \quad (5)$$

To generate SPSS optimisation key factors into requirements, related domain knowledge is required. Domain knowledge could be collected from interview with domain expertise, work logs from operators, requirements specification documents, etc. In order to optimise information retrieval and utilisation, knowledge graphs are used to manage domain knowledge, which could lay the foundation for advanced analytics, decision support, and strategic planning. Moreover, the inherent ability of knowledge graphs to encode both explicit and implicit relationships enables nuanced knowledge retrieval and inference, thereby enhancing the efficiency and depth of knowledge discovery processes. With massive, collected knowledge, the knowledge graph $G = \langle V, E \rangle$ about SPSS-related knowledge could be constructed, in which V represents the set of nodes in KG, and E represents a collection of relations between nodes of it. These key structures can be extracted from the original document according to the pre-defined ontology concept. Large amounts of product modules, service modules, context, fault messages, etc., could be included in the SPSS-related domain knowledge graph. For some popular areas with some reliable KG that could be found conveniently, it is encouraged to directly use well-constructed and mature graphs to assist this part of the work.

With well-constructed knowledge graphs, valuable information could be obtained with the help of graph query technology. Key factors generated from section 3.3 will be used as the query input, which will be transferred to complete optimisation requirements. It's possible that some useless information or even nothing could be queried out from some requirement factors, which may be due to the accuracy of the model or the quality of data. Moreover, in the process of transforming requirement factors to requirements, it is encouraged to get some advice from domain experts if necessary.

4. Case study

In order to test and discuss the proposed cognitive intelligence-enabled requirements elicitation framework, wind turbines are used as objects of the case study. In recent years, clean energy, specifically wind energy, has become increasingly important as a power source and has been widely adopted. However, as a type of large-scale equipment that requires long-term unattended operation of wind turbines, how to get the design requirements of different contexts has become a growing concern for designers. This study is based on the digital twin-based data collection method and the wind turbine knowledge graph.

4.1. Digital twin-based data collection

Wind turbines are composed of major subsystems containing rotors, drive train system, electrical system, control system, etc. As shown in Figure 6, to establish the digital twin-based data collection process proposed in section 3.1, the first step is to install proper sensors for important components. 18 major components including rotor, nacelle, gearbox, etc. have been placed with sensors such as wind sensors, temperature sensors, vibration sensors, etc. While SPSS working, massive real-time data could be collected through sensors and transmitted with industrial Modbus, Bluetooth, etc. Since there are often problems such as missing and duplicated data in the original data, necessary data processing

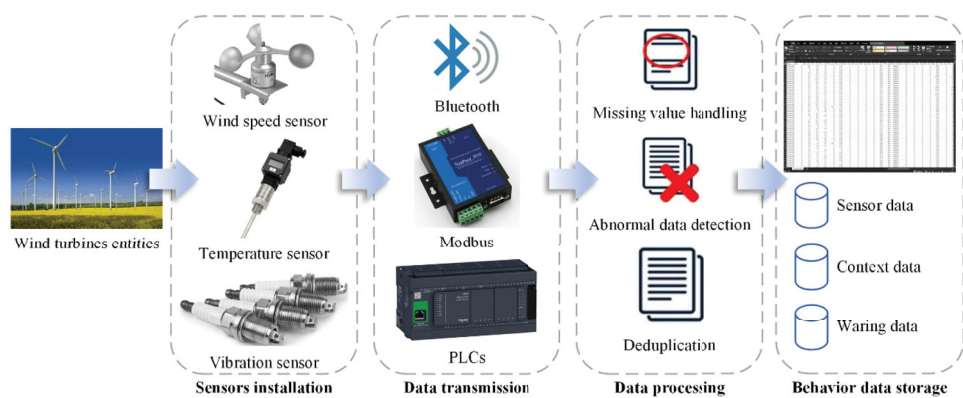


Figure 6. Digital twin-based data collection process for wind turbines.

Table 2. Result of data collection and division of dataset.

	Content	Data volume
Training set	normal state	2800
Validation set	normal state & abnormal state	299
Test set	normal state & abnormal state	299

methods should be adopted, concluding missing data handling, abnormal data detection, deduplication, etc.

Moreover, Digital twin-driven data collection process gathered sensor data, context data, and warning data together into the twin data. 18 kinds of sensor data are collected including parameters from the key components of wind turbines such as inverter temperature, cabinet temperature, nacelle ambient temperature, main carrier temperature, etc. 2 kinds of context data including wind speed and environment temperature. Warning data mainly represents whether the working state is normal or not.

During the process of data processing, eigenvalues are normalised and shrank into the range of [0, 1]. Part of the data is randomly extracted from the normal data sample of the wind turbine with a particular proportion, and this part of the data is divided into two halves of an equal number, which are used as the verification set and the test set. The remaining normal data is substituted into the model training as the training data set. To validate the abnormal data of the wind turbine, it is randomly divided into two equal halves. Furthermore, some of the previously extracted abnormal state data are combined to form verification and test sets. The result of the division of the dataset is shown in Table 2.

4.2. Behaviour modelling-based cognition of wind turbines

According to the data characteristics of the dataset, this study uses an AE model for training to describe wind turbines' normal behaviour. To identify the potential characteristics of the monitoring object, the number of nodes in the hidden layer is lower than that of the input and output layers. Additionally, the number of nodes in the input and output layers matches the number of dimensions of the wind turbine data sample. Eighteen features are selected, representing sensor data from wind turbines' main components. The activation

Table 3. The structure of the AE network.

Number	Layer	Number of nodes
1	Input layer	18
2	Hidden layer_1	8
3	Hidden layer_2	4
4	Hidden layer_3	8
5	Output layer	18

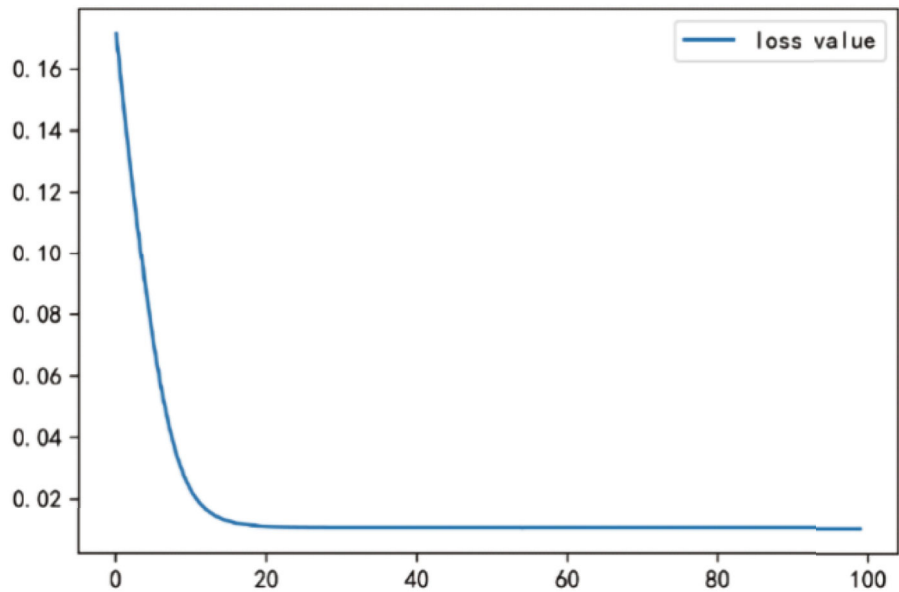


Figure 7. Loss value of AE model.

function used in the hidden layers is the linear rectification function (ReLU). Since the operational data of wind turbines use inter-zone scaling, the tanh activation function is used in the output layer of the AE model. The structure of the AE network is shown in Table 3.

Because the training goal is to make the reconstructed input closer to the original input as much as possible, the reconstruction error is expected to be small enough. To make the trained model more robust, converge faster during training, and extract valuable patterns from noisy data effectively, smooth L1 is used as the loss function in this study. The smooth L1 loss function combines the MAE loss function and the MSE loss function, which reserves both advantages. The Adam optimiser is also applied during the training phase to solve the learning problem. 200 training rounds are selected according to the learning effect, and the loss value change is shown in Figure 7.

After the model training is completed, the data in the test set is substituted into it. The reconstruction error between the wind turbine’s normal and abnormal state data could be calculated. The training effect of the model is whether the two states could be distinguished effectively, as shown in Figure 8. It can be seen clearly that the normal and abnormal states are distributed in different ranges, and the reconstruction error of the abnormal state is obviously larger than the normal state. We can draw the conclusion that the AE model

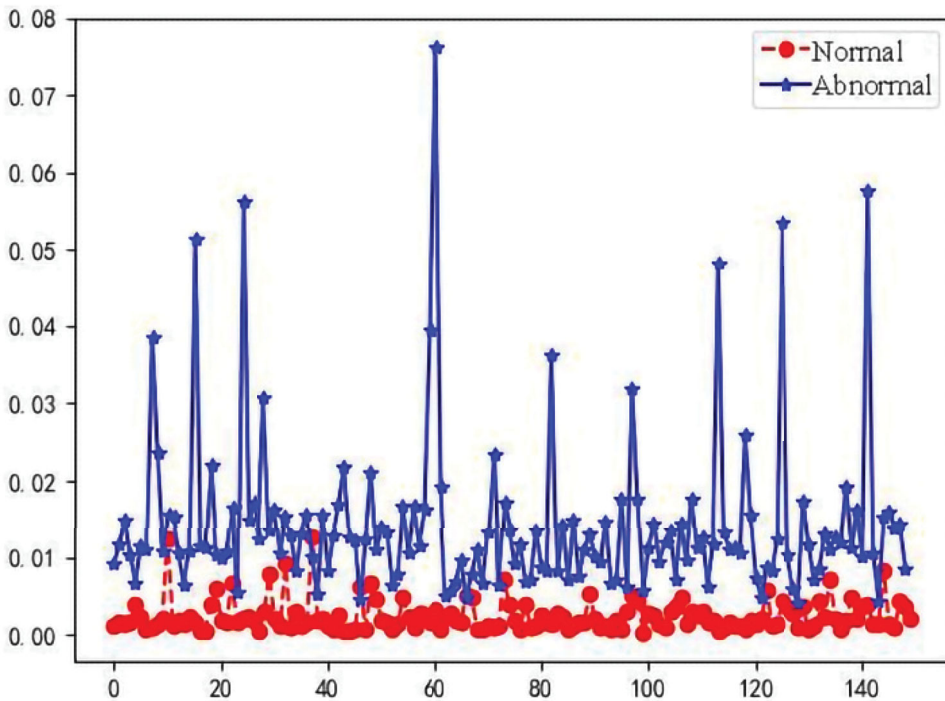


Figure 8. The reconstruction error between the normal and abnormal state data.

designed could model the normal behaviour of the wind turbines, and the following experiment about model interpretation could be carried out well on the basis of the stable and accurate model.

4.3. Model interpretation

The key structural parameters of the abnormal state of SPSS are input to the well-trained behaviour model. An abnormal state would be judged according to the reconstruction error, which is larger than the normal situation. Therefore, exploring how to find input parameters that significantly impact model state judgment in different contexts can be regarded as a key structure with a more significant impact on the abnormal state through data science methods and can be further transformed into related implicit design requirements.

To improve the accuracy of analysis, data is divided and classified according to different contexts. According to the suggestions of experts in the field of wind power and combined with the data characteristics, four kinds of context have been taken into consideration. These four kinds of context are composed of high ambient temperature, low ambient temperature, high wind speed, and low wind speed.

In the process of interpretability analysis of the model, the model interpretation method, which is independent of the model, is adopted. Firstly, based on the correlation theory in the union game theory, the average marginal contribution of different feature values is characterised by calculating the shapley value of each one. The larger the shapley value is, the

Table 4. Optimisation key factors.

Context	Number of the optimisation key factor	Component	Mean(Shapley value)
High wind speed	S11	Inverter	0.01061151
	S12	Nacelle	0.00622623
	S13	Control cabinet	0.00561981
	S14	Rotor	0.00511454
	S15	Yaw inverter	0.00476214
Low wind speed	S21	Rotor	0.01669337
	S22	Rear bearing	0.01109223
	S23	Stator	0.0064117
	S24	Fan inverter	0.00630861
	S25	Spinner	0.0037841
High temperature	S31	Stator	0.00709699
	S32	Rotor	0.00235949
	S33	Inverter	0.00227895
	S34	Rectifier	0.00210477
	S35	Fan inverter	0.00171834
Low temperature	S41	Rotor	0.007977
	S42	Stator	0.00474579
	S43	Control cabinet	0.00406751
	S44	Transformer	0.00403209
	S45	Tower	0.00341251

greater the influence of this part of the component on the usage state of the wind turbines under its related context.

The model interpretability analysis is carried out separately for the AE models of four kinds of context. Summarising the relevant factors in the above four kinds of context, optimisation key factors could be transformed from the model interpretation results, which could be used to guide the following design process. The first five components with the most significant impact in each context are selected as the optimisation key factors for a total of 20. As Table 4 shows.

4.4. Domain knowledge graph aided SPSS optimisation requirements generation

The wind turbine domain knowledge graph used in this experiment is from XAI4Wind (Chatterjee and Dethlefs 2020). The XAI4Wind mainly contains the alarm information during the operation of the wind turbines, error correction information, fault diagnosis information, and preventive measures. As shown in Figure 9, XAI4Wind is stored in the Neo4j database, where the query statements in the cypher syntax are used to input the context and component elements in the key factors into the map for querying, combined with domain knowledge for screening.

For example, it can be found that after searching for the context keyword of 'high temperature' and the component keyword of 'Transformer', the relevant nodes that appear are 'monitoring operating parameters' and 'opening the ventilation fan under the transformer', which can be converted into 'in the case of high temperature, the operating parameters of the transformer should be monitored in real-time' and 'in the case of high temperature, Transformers should be given adequate ventilation' and other demand regulations. For the optimisation key factors, as stated in Table 4, the requirements strip obtained after completion and transformation in XAI4Wind is shown in Table 5. It is worth noting that because the knowledge graph is challenging to cover comprehensive knowledge of the fan field, it may

Table 5. SPSS optimisation requirements.

Context	SPSS key factor		SPSS optimisation requirement
High wind speed	S12	S12.1	< high wind speed, cabin, inspection of cables, automatically >
		S12.2	< high wind speed, cabin, lubrication, automatically >
	S13	S13.1	< high wind speed, cabin, pay attention to the internal temperature, continuously >
		S13.2	< high wind speed, cabin, check wiring conditions, continuously >
	S14	S14.1	< high wind speed, rotor, check wiring conditions, continuously >
		S14.2	< high wind speed, rotor, increase the awareness of the environment, automatically >
		S14.3	< high wind speed, rotor, check wiring conditions, more frequently >
Low wind speed	S22	S22.1	< low wind speed, bearing, realise monitoring of its own running noise to reflect the running state, real-time >
		S22.2	< low wind speed, bearing, be kept lubricated, continuously and effectively >
High temperature	S32	S32.1	< high temperature, rotor, maintain lubricated, continuously and effectively >
		S32.2	< high temperature, rotor, be monitored for its running noise, continuously >
Low temperature	S44	S44.1	< low temperature, transformer, pay attention to heat dissipation, continuously >
		S44.2	< low temperature, transformer, be inspected, continuously and effectively >
	S45	S45.1	< low temperature, tower, be enhanced to avoid vibration, effectively >

specific requirements, it is feasible and low-carbon to do this process with the help of existing physical products and services. It is recommended that each piece of SPSS optimisation requirements should be taken into consideration whether it could be met within existing resources firstly. Two ways for meeting SPSS optimisation requirements are shown as follows.

- (1) Satisfy the optimisation requirements through service optimisation. For example, maintenance services could be set to fulfil the requirements, such as breakdown maintenance or daily inspection. Some development on software could be added to realise more humanised functions.
- (2) Satisfy the optimisation requirements by product iteration of SPSS. Design flaws always exist in the actual product, and more advanced technologies could be used to develop the operating performance. For this part of optimisation requirements, redesigning a new generation of SPSS is encouraged to provide greater market competitiveness for enterprises.

With the comprehensive management of the existing resources and the suggestions from experts in related fields, the optimisation solutions for optimisation requirements in Table 5 are shown in the last two columns of Table 6.

Five experts engaged in wind turbine design, operation and maintenance were invited to evaluate the assistance role of the proposed framework in actual engineering. They were asked to express their opinions on the design process using the traditional requirements elicitation method (method A in Table 7) and the design process assisted by the framework

Table 6. Solutions according to the elicited optimisation requirements.

Optimisation requirements	SPSS optimisation design solutions:	
	Service optimisation	Product iteration
S12.1	• Increase manpower for inspection. • Increase the frequency of inspections.	
S12.2	• Increase the frequency of lubrication system inspections.	• Design a more reasonable lubrication system.
S13.1		• Install sensors for detecting the internal temperature. • Set temperature alarm.
S13.2	• Arrange personnel to inspect the wiring.	
S14.1	• Arrange personnel to inspect the wiring.	
S14.2		• Deploy environmental sensors. • Apply rotor control algorithm based on environmental information
S14.3	• Arrange personnel to inspect the wiring.	
S22.1		• Deploy noise sensors. • Apply Noise-Based Monitoring Algorithms for Running Turntables
S22.2	• Increase the frequency of lubrication system inspections.	• Design a more reasonable lubrication system. • Install sensors for lubricating oil detection.
S32.1	• Increase the frequency of lubrication system inspections.	• Design a more reasonable lubrication system. Install sensors for lubricating oil detection.
S32.2		• Deploy noise sensors.
S44.1		• Deploy temperature sensors. • Increase efficient cooling device.
S44.2	• Regular inspection.	
S45.1		• Apply a more rigid tower structure. Vibration sensors installed on the tower body.

Table 7. Evaluation results from five experts.

Method	Immediacy					Accuracy					Impact on engineering design process				
A	7	6	5	8	6	5	7	6	4	5	6	5	4	6	4
B	8	9	7	9	8	9	8	9	7	8	7	8	8	8	7

proposed in this paper (method B in Table 7) by filling in a questionnaire, from the three aspects of immediacy, accuracy of the elicited requirements and impact on engineering design process. For each aspect, score 1 represents the worst, 10 represents the best, and only integer values could be chosen. The evaluation results are shown in Table 7.

The comparative analysis reveals that the proposed framework effectively facilitates the optimisation process for SPSS design. Application of this framework to an industrial case study, followed by a survey involving domain experts, demonstrates its suitability for optimising SPSS design. The framework addresses the optimisation requirements of SPSS design by leveraging real-time data during the usage stage of the SPSS lifecycle. This enables service innovation during usage and facilitates product iteration during the design phase of the next generation. Moreover, the framework integrates cognitive intelligence to uncover vital information within the vast datasets generated during SPSS usage, bridging the gap between usage and design stages. With the growing structural and functional

complexity, the proposed framework offers potential cost and complexity reductions in SPSS design optimisation endeavours. The framework's flexible nature enables it to evolve alongside technological advancements and changing user needs, ensuring its relevance and effectiveness in the face of dynamic environments.

5. Conclusion

SPSS is a newly appeared business model that is more intelligent and complex than traditional products. Through its smart interconnection function, SPSS could realise the real-time collection and transmission of data. The potential value of data can now be excavated with the assistance of cognitive intelligence based on machine learning and deep learning, such as personalised requirements for product design. In this way, through the cognitive intelligence-enabled process with data and knowledge, it is feasible and meaningful to get the optimisation requirements for the design of SPSS.

This study suggests a new cognitive intelligence-enabled approach to generating SPSS optimisation requirements by analysing usage data. To realise SPSS cognition, the behaviour model is constructed based on the Auto-Encoder model which belongs to deep learning technologies. With data collected through the digital twin technology of the usage stage, the cognition model could depict the working status of SPSS. Moreover, a model interpretation-based SPSS cognition intelligence architecture is proposed. Through SHAP, a model explanation method, cognitive intelligence analysis is carried out to explore SPSS's key components strongly related to the typical context. Finally, in order to effectively support the design optimisation process, an improved optimisation requirement template is given, which could be fulfilled with the assistance of a domain knowledge graph. Furthermore, this study suggests a framework and conducts a case study of wind turbines.

Nonetheless, the authors hope this study could provide ideas and inspiration for cognitive intelligence-enabled SPSS optimisation design. At the same time, the following directions are worth considering in the future. (1) Improve the requirement expression template and consider the implementation of multi-model process elements. (2) The requirements generation process still depends on the knowledge and judgment of experts in related fields. In the future, we could consider using deep learning or reinforcement learning algorithms to achieve a more automatic and intelligent process. (3) In the actual industrial scenarios, the performance of SPSS is often affected by external factors such as environmental conditions, user operating proficiency, etc. How to reduce the interference of unexpected external factors and improve the robustness of the framework during the design optimisation requirement elicitation process deserves attention.

Disclosure statement

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